

Positivity of the leading coefficient for a nonnegative face amplitude

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Abstract

If $\eta(0, \cdot) \geq 0$ on the closed noncritical box and $\eta(0, v_1) \neq 0$ at some point of the closed box, continuity produces an interior point where $\eta(0, \cdot) > 0$ (the closed box is the closure of the open box), so the leading coefficient is positive and the asymptotic equivalence $Z \sim CN^{-p} \log^{d_0} N$ holds. This is the sharp form of the positivity hypothesis recommended in the third Astra consult: nonnegativity on the minimal face, not on the whole box (already $\eta(u) = u$ in $d = 1$ changes the leading exponent). Zero `sorry/axiom`.

1 The things formalised

```
theorem exists_interior_pos {d' : } (b : ) (hb : 0 < b) (e : (Fin d' → ) → )
  (hec : Continuous e) (v : Fin d' → ) (hv : i, 0 < v i < v i b)
  (hpos : 0 < e v) :
  v : Fin d' → , ( i, 0 < v i < v i b) 0 < e v
theorem blockCoeff_pos_of_nonneg_ne_zero ...
  (henn : v : Fin d' → , ( i, 0 < v i < v i b) → 0 < 0 v)
  (v : Fin d' → ) (hv : i, 0 < v i < v i b) (hne : 0 v < 0) :
  0 < blockCoeff b p k k' h'
theorem blockStateIntegral_isEquivalent_of_ne_zero ... :
  (fun N : => blockStateIntegral b N k h k' h' ) ~[atTop]
  fun N : => blockCoeff b p k k' h' * (N ^ (-p) * Real.log N ^ d)
```

2 Proof strategy

`closure_pi_set` and `closure_Ioo` identify the closed box as the closure of the open box; `mem_closure_iff` applied to the open set $\{e > 0\}$ yields an interior point with $e > 0$, and unit 90's positivity theorem finishes.

3 Outcome

Astra #3 target 1 is complete. Next: the polynomial jet algebra for $\eta e^{\beta s \xi}$ and the all-order one-dimensional theorem with $c_j = k^{-1} \sum_{\ell} p_{j,\ell} S_{q_j+\ell}(a)$.